

Unsupervised Learning 2025/2026

Project Report: Milestone 2

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Field	Value
Project title	Behavioural Segmentation of Travel Agency / Tour Operator Bookings in the Hotel/Resort Industry
Checkpoint date	30 April 2026
Repository	https://github.com/Leonorafonso28/hotel_booking_demand

1. RQ1 and segmentation time

- **RQ1.** Among bookings made through Travel Agency / Tour Operator (TA/TO) intermediaries, are there latent behavioural groups characterised jointly by booking timing, stay design, party composition and commercial-channel metadata that are observable at booking time?
- **Unit of analysis:** one row = one booking record.
- **Segmentation time (index time):** immediately after the booking is recorded (booking-time). A variable is admissible as a clustering input only if its value is known at this moment. As a direct consequence, *booking_changes*, *assigned_room_type*, *days_in_waiting_list*, *is_canceled* and *reservation_status* are excluded from clustering inputs and used only as post-hoc profilers.

2. Dataset version control and documentation

- **Dataset release:** Hotel Booking Demand (course release for 2025/2026); 119,390 booking records, 32 columns; arrivals 1 Jul 2015 - 31 Aug 2017.
- **File fingerprint (MD5):** 5bf588c5a949443e021fb7c847d31b27 - recorded in the notebook at load time and printed for traceability.
- **Source / reference / usage terms:** António, N., de Almeida, A., & Nunes, L. (2019). Hotel booking demand datasets. Data in Brief, 22, 41–49. <https://doi.org/10.1016/j.dib.2018.11.126>. License CC BY 4.0.
- **Known data-quality issues identified before modelling:**
 - *country* missing in 0.14% of TA/TO rows; imputed to "PRT" per the reference paper's domestic-default convention.
 - *children* has 4 NaNs across the full dataset; imputed to 0.
 - *agent* is sparse (3.2% missing) and *company* is very sparse (99.1% missing); both are id-like high-cardinality fields and excluded from clustering on principle.
 - Records with adults+children+babies = 0 are not valid behavioural units (no guests); dropped before modelling (120 rows).
 - *ADR* has a small number of records with $adr \leq 0$ (complimentary/test bookings) and one record with $adr \sim 5400$ (data-entry error); these are filtered when ADR is used (post-hoc profiling and the withADR sensitivity variant only).

3. Completed Exploratory Data Analysis (EDA)

Variables dropped before modelling

- *agent*, *company*: id-like high-cardinality / sparse fields. Including them would over-determine clusters and break the privacy assumption that intermediary identifiers are not used.
- *arrival_date_year*: only 2015–2017; would split clusters by data-collection span, not behaviour.
- *arrival_date_week_number*, *arrival_date_day_of_month*: redundant with *arrival_date_month* at the granularity needed for behavioural segmentation.
- *stays_in_week_nights*, *stays_in_weekend_nights*, *adults*, *children*, *babies*: absorbed into engineered features *total_nights*, *weekend_share*, *party_size*.

Mandatory leakage exclusions

- *is_canceled, reservation_status, reservation_status_date*: outcome / post-event variables, not available at booking time. Excluded from clustering and used only as post-hoc profilers.
- *booking_changes, assigned_room_type, days_in_waiting_list*: post-decision variables (updated by hotel operations after booking). Excluded from clustering on the booking-time index defined in §1; *booking_changes* is reported only in post-hoc profiling.

4. Main representation (RQ2)

Variables included as clustering inputs

- **Numerical (8)**: *lead_time, total_nights, weekend_share, party_size, previous_cancellations, previous_bookings_not_canceled, required_car_parking_spaces, total_of_special_requests*.
- **Binary (1)**: *is_repeated_guest* — kept as 0/1, NOT scaled (matches the one-hot block scale).
- **Categorical (7)**: *market_segment, reserved_room_type, meal, customer_type, deposit_type, country, arrival_date_month*.

Excluded / profiling-only variables

- *hotel* (Resort vs City): kept as profiler so we can verify clusters are not hotel-driven.
- *booking_changes*: post-decision under booking-time index; profile-only.
- *adr*: price/revenue outcome. Excluded from R0 (cluster-forming variables prioritise behavioural booking descriptors; ADR is reserved for post-hoc interpretation). Used only in post-hoc profiling and in the withADR sensitivity variant (R2).
- *distribution_channel*: constant within R0 (all rows are TA/TO). Reassessed in the R0-alt comparison on the parent dataset.

Numerical preprocessing

- **Imputation**: median for numerical, most-frequent for binary, most-frequent for categorical.
- **Capping**: *previous_cancellations* and *previous_bookings_not_canceled* capped at P99; *required_car_parking_spaces* capped at 3; *total_of_special_requests* capped at 5.
- **Transforms**: none. (Log scale is for real-valued continuous data, not all numerical data.) $\log_{1p}$ was removed from the count features; *lead_time* is kept on its native scale in R0 and a $\log_{1p}(\text{lead_time})$ variant is registered as R4 for the sensitivity comparison.
- **Scaling**: StandardScaler on the 8 numerical inputs; no scaling on the binary and one-hot blocks (those are already 0/1).

Categorical preprocessing

- **Rare-category rule**: levels with frequency < 1% are grouped under "Other". On TA/TO this groups, e.g., 4 rare market segments and 160 rare countries.
- **Encoding**: `OneHotEncoder(handle_unknown='ignore', sparse_output=False)` — unknown levels at inference time map to all-zero rows in the encoded block (no error, no leakage).
- **Final encoded matrix**: 97,750 rows $\times$ 57 columns (8 numerical + 1 binary + 48 one-hot).

Final representation_id

R0-standard-noLog-noADR-marketSegment-countryRare1pct

5. Baseline modelling evidence

Methods and parameter ranges

- **k-means**: `init='k-means++', n_init=5, max_iter=300, random_state=12345`.
- **iK-Means**: Mirkin's Iterative Anomalous-Pattern (AP) procedure with `min_cluster_size=50` on a fixed-seeded subsample of 20,000 records;
- **Fixed K-grid**: $k \in \{2, 3, 4, 5, 6, 7, 8\}$. Lower bound: any meaningful partition. Upper bound: interpretability ceiling for a behavioural segmentation report.

- **iK-Means stop criteria:** inner loop terminates on (a) membership stability ($S_t = S_{t-1}$), (b) centroid stability ($\|c_{new} - c_{old}\| \leq 1e-9$), or (c) the hard cap $max_iter = 300$. Outer loop terminates when the residual set is empty; AP clusters with size < 50 are discarded as initial seeds (kept in the AP record for traceability).

Internal validity indices (computed in the same R0 Euclidean space)

k	Inertia	Silhouette	Calinski-Harabasz	Davies-Bouldin	K-Means (s)	iK (s)
2	943,187	0.116	11,607	2.624	1.09	0.22
3	847,042	0.129	12,010	2.253	1.10	0.30
4	754,969	0.140	12,956	1.921	1.68	0.58
5	704,930	0.129	12,141	1.985	1.71	0.53
6	657,524	0.139	11,823	1.820	1.88	0.27
7	619,026	0.135	11,478	1.806	1.68	0.20
8	586,004	0.142	11,179	1.730	2.04	0.36

Bold green = winning value of that index. $k = 8$ is the Silhouette argmax and the Davies-Bouldin argmin; $k = 4$ is the Calinski-Harabasz argmax (highlighted yellow) and the elbow of the Inertia curve.

Selection rule applied: Silhouette + elbow + parsimony tiebreak

Silhouette max: $k = 8$ (0.142);

Elbow: $k = 4$ (largest second difference; WCSS flattens after 4; **CH** also peaks here);

Silhouette gap ($k = 4$ vs 8): 0.0018 (very small difference).

Selected $k = 4$ (parsimony). Supporting metrics at $k = 4$: CH = 12,956 (highest), DB = 1.921 (lower than $k = 2, 3, 5$). If it were for major voting, the selected k would be 8, but since the difference in the metrics at $k=4$ and $k=8$ are small for the ones where $k=8$ is better, we've chosen the smaller k .

First cluster-interpretation note ($k = 4$, in original units)

Each cluster's identity is summarised below using the most discriminating feature in the % deviation from grand mean table:

Cluster	Size (% of TA/TO)	Identity (largest deviation from grand mean)	Median lead_time / nights	Cancellation rate (post-hoc)
0	4.6% (4,478)	Parking-required road-trippers (+2,083% on parking_spaces; +60% special requests)	31 d / 3 n	0.0%
1	50.6% (49,498)	Plain offline-TA bookings (-81% special requests; below average on most engagement features)	97 d / 3 n	46.1%
2	39.2% (38,300)	Engaged online-TA bookings (+109% special requests; +14% total_nights, +15% party_size)	64 d / 3 n	31.1%
3	5.6% (5,474)	Very early group bookers with cancellation history (+92% lead_time; +1,685% previous_cancellations)	260 d / 2 n	99.2%

Interpretation. Cluster 1 and Cluster 2 together account for ~90% of the TA/TO sub-population and differ chiefly on engagement (special-request load, party size, stay length); Cluster 0 and Cluster 3 are smaller, sharper minorities at opposite ends of the cancellation spectrum.

6. Sensitivity / robustness plan already defined

- **Main representation:** R0-standard-noLog-noADR-marketSegment-countryRare1pct (8 numerical + 1 binary + 48 one-hot, Euclidean metric, no log).
- **Resampling rule:** two complementary protocols, both with at least 10 runs.
 - **Random restart** - 5 seeds: [12345, 23456, 34567, 45678, 56789], k-means refit on R0 for each. Captures the lottery of random initialisation.
 - **Bootstrap resampling** - B = 20 bootstrap samples drawn with replacement from R0 (full size). K-means fitted on each. Captures stability under data perturbation. Already executed for k = 4 (see below).
- **Stability measure:** pairwise Adjusted Rand Index (ARI) on the intersection of common rows; reported as mean $\pm$ std and min/max. Mean ARI > 0.75 = stable; 0.6–0.75 = moderately stable; < 0.6 = unstable.

Bootstrap stability already executed (k = 4, R0)

Method $\times$ k $\times$ B	Value
KMeans $\times$ 4 $\times$ 20 (190 pairwise ARIs)	
Mean pairwise ARI	0.644
Std / min / max ARI	0.310 / 0.139 / 0.996
Verdict	Moderately stable: the 4-cluster structure holds under bootstrap resampling, though some resamples merge two large clusters (min ARI = 0.14). This suggests k = 4 captures the main structure: Clusters 0 and 3 are robust, while the boundary between Clusters 1 and 2 (together ~90% of the sub-population) is diffuse.

Sensitivity question and predeclared variants

Each variant differs from R0 in exactly one preprocessing decision (the lecturer's recommended controlled comparison rule). The question each variant answers is stated up front.

Variant	One-decision change vs R0	Question answered
R0-alt	distribution_channel of the parent dataset replaces market_segment	Are clusters driven by the channel block?
R1	RobustScaler instead of StandardScaler	Are clusters driven by outliers / heavy tails?
R2	ADR included as scaled numerical input	Do clusters become price/value segments?
R3	country dropped from inputs	Are clusters mainly geographic?
R4	log1p(lead_time) replaces lead_time	Does a log transform of lead_time change the partition?

7. Experiment logging and repository status

- **experiments.csv exists.** With 14 rows from Task 1 (7 KMeans runs + 7 iK-Means runs over the K-grid). Fixed schema: date, run_id, representation_id, method, parameters, seed, sample_rule, k, silhouette, calinski_harabasz, davies_bouldin, inertia, time_s, notes.
- **Environment:** requirements.txt exists at the repo root with pinned major versions (numpy $\geq$ 1.26, pandas $\geq$ 2.0, scikit-learn $\geq$ 1.4, matplotlib $\geq$ 3.7, seaborn $\geq$ 0.13, jupyter, nbformat). An environment.yml can be added on request — the dependencies are pure Python scientific stack with no platform-specific pinning needed.
- **Run command / entry point:** the notebook notebooks/ANS_Task1_revised_v2.ipynb reproduces the entire pipeline end-to-end when executed in Jupyter. It locates the project root automatically (walks up the directory tree until it finds requirements.txt) and writes outputs into results/figures/, results/tables/ and results/reports/experiments.csv.

Appendix

ColumnTransformer / Pipeline sketch

Transformer Name	Input Features	Pipeline Steps	Transformation Logic
num	Numerical	SimpleImputer StandardScaler	⇒ Impute missing values with median; scale to zero mean and unit variance.
num_bin	Binary	SimpleImputer	Impute missing values with most frequent; maintained as binary 0/1.
cat	Categorical	SimpleImputer OneHotEncoder	⇒ Impute with most frequent; encode categorical labels into one-hot binary vectors.

Missingness summary (TA/TO sub-population, by attribute)

Attribute	Type	Missing (%)	Treatment
company	identifier	99.07%	excluded (id-like, sparse)
agent	identifier	3.24%	excluded (id-like, high-cardinality)
country	categorical	0.14%	imputed "PRT" (domestic default)
children	count	< 0.01%	imputed 0

Outlier summary (relevant numerical variables)

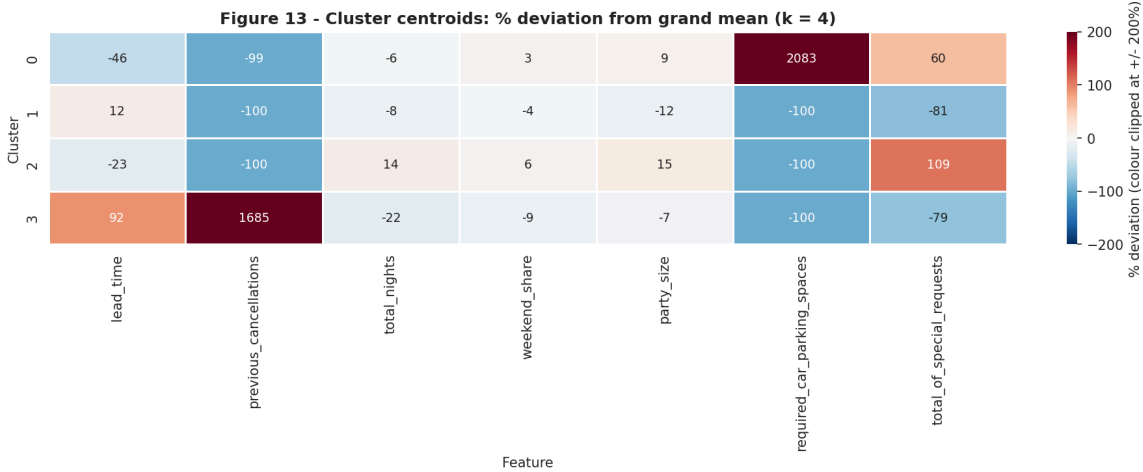
Variable	Issue	Treatment
lead_time	right-skew (skew=1.26); P99 ~ 370 days	StandardScaler only in R0 (no log); log variant tested in R4 sensitivity
previous_cancellations	small count, mostly 0; max 26	cap at P99 + StandardScaler; NO log (counts, not real-valued)
previous_bookings_not_canceled	small count, mostly 0	cap at P99 + StandardScaler; NO log
required_car_parking_spaces	> 90% are 0; max 8 (likely error)	cap at 3 (domain ceiling) + StandardScaler
total_of_special_requests	count 0–5 dominant; max 5 already	cap at 5 + StandardScaler
adr	non-positive (complimentary) and one extreme (~5,400)	EXCLUDED from clustering inputs; filtered (0 < adr ≤ 1000) only for post-hoc profiling and R2 sensitivity

Experiments.csv excerpt (baseline-results table — k-means and iK-means)

date	run_id	method	parameters	seed	k	silhouette	time_s
2026-04-30	kmeans-k2-seed12345	KMeans	k-means++, n_init=5, max_iter=300	12345	2	0.116	1.09
2026-04-30	kmeans-k4-seed12345	KMeans	k-means++, n_init=5, max_iter=300	12345	4	0.140	1.68
2026-04-30	kmeans-k8-seed12345	KMeans	k-means++, n_init=5, max_iter=300	12345	8	0.142	2.04
2026-04-30	ikmeans-k4-seed12345	iKMeans-AP	AP_min_size=50, AP_subsample=20000	12345	4	0.140	0.58

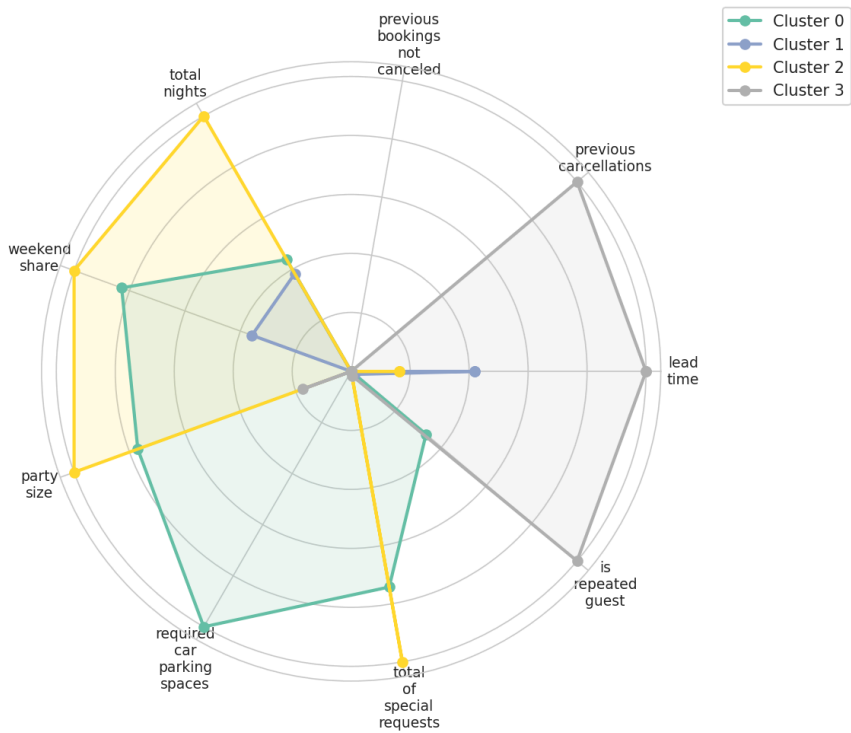
date	run_id	method	parameters	seed	k	silhouette	time_s
2026-04-30	ikmeans-k8-seed12345	iKMeans-AP	AP_min_size=50, AP_subsample=20000	12345	8	0.142	0.36

Initial cluster-profile figures (k = 4)



Cluster centroids: % deviation from the grand mean (k = 4). Each row is a cluster, each column a numerical clustering input; the colour scale is clipped at +/- 200% so extreme cells (e.g. parking_spaces in Cluster 0, previous_cancellations in Cluster 3) do not wash out the rest.

Figure 9 - Cluster profiles (numerical inputs), k = 4



Radar of the four cluster profiles in normalised numerical inputs (each axis rescaled to [0, 1] across cluster means). Cluster 0 stands out on parking_spaces and special_requests; Cluster 3 on lead_time and previous_cancellations; Clusters 1 and 2 occupy the broad middle of the space and differ chiefly on engagement (special_requests, total_nights, party_size).